

Bayesian Optimization of Sensor Positions in Magnetic Detection Electrical Impedance Tomography

Theory companion to `studies/optimal_sensors_bayesian_approach/main.m`

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Abstract

This document describes the theory implemented in `main.m` for optimizing the positions of magnetometers in magnetic detection electrical impedance tomography (MDEIT). The approach follows the Bayesian optimal experiment design framework of Hyvönen, Seppänen and Staboulis [*Optimizing electrode positions in electrical impedance tomography*, SIAM J. Appl. Math., arXiv:1404.7300], who optimized *electrode* positions in EIT. Here the design variables are instead the positions of external magnetic field sensors. The complete electrode model is linearized around the prior mean conductivity, which yields an explicit Gaussian posterior covariance. The sensor positions are then chosen to minimize either the trace (A-optimality) or the log-determinant (D-optimality) of the posterior covariance, using analytic gradients obtained from adjoint solutions and explicit derivatives of the Biot–Savart kernel.

1 Problem setting

Let $\Omega \subset \mathbb{R}^3$ be a conducting body with electrical conductivity σ , and let $M \geq 1$ magnetometers be located at positions $\mathbf{s}_m \in \mathbb{R}^3 \setminus \overline{\Omega}$, $m = 1, \dots, M$. Currents are injected through L fixed surface electrodes using N stimulation patterns. Each injected pattern drives an internal current density $\mathbf{J} = -\sigma \nabla u$, whose magnetic flux density is measured by the sensors.

The inverse problem of MDEIT is to recover the (discretized) conductivity $\sigma \in \mathbb{R}^{n_{\text{el}}}$ from these magnetic measurements. The *design* problem considered here is: given a prior distribution for σ and a noise model, place the sensors $\{\mathbf{s}_m\}$ so that the posterior distribution of σ given the data is as localized (informative) as possible.

Two features distinguish this setting from the EIT electrode-position problem of Hyvönen et al.:

1. The electrodes are *fixed*; only the contact-free magnetometers move. Consequently the forward PDE and its finite element (FE) system matrix do not depend on the design variables.
2. The sensor positions enter the measurement map only through the smooth Biot–Savart kernel. The derivatives with respect to the design variables are therefore ordinary (explicit) derivatives of a volume integral; no shape derivatives of the boundary value problem are required. This is analytically and computationally simpler than the electrode-position case, which requires second-order Fréchet (shape) derivatives of the complete electrode model.

2 Forward model

2.1 Complete electrode model

The electric potential u inside Ω and the electrode potentials $U = [U_1, \dots, U_L]^\top$ satisfy the complete electrode model (CEM):

$$\begin{aligned} \nabla \cdot (\sigma \nabla u) &= 0 && \text{in } \Omega, \\ \sigma \partial_\nu u &= 0 && \text{on } \partial\Omega \setminus \cup_e E_e, \\ u + z_e \sigma \partial_\nu u &= U_e && \text{on } E_e, \quad e = 1, \dots, L, \\ \int_{E_e} \sigma \partial_\nu u \, dS &= I_e, && e = 1, \dots, L, \end{aligned} \quad (1)$$

where z_e are the contact impedances and $I = [I_1, \dots, I_L]^\top$ the injected current pattern (with $\sum_e I_e = 0$). Discretizing with piecewise linear finite elements yields the symmetric block system

$$\begin{pmatrix} A_c & A_e \\ A_e^\top & A_d \end{pmatrix} \begin{pmatrix} u \\ U \end{pmatrix} = \begin{pmatrix} 0 \\ I \end{pmatrix}, \quad (2)$$

where $A_c \in \mathbb{R}^{n_n \times n_n}$ contains the σ -weighted stiffness matrix plus contact terms, and A_e, A_d contain only contact-impedance terms (independent of σ). Eliminating U gives the Schur complement

$$S(\sigma) = A_c - A_e A_d^{-1} A_e^\top, \quad (3)$$

which is the symmetric positive (semi)definite operator used for all forward and adjoint solves (function `M` in the code). Only A_c depends on σ , so

$$v^\top \frac{\partial S}{\partial \sigma_k} w = \int_{T_k} \nabla v \cdot \nabla w \, dV = |T_k| (\nabla v)_k \cdot (\nabla w)_k, \quad (4)$$

for any FE functions v, w , where T_k is the k -th tetrahedron and the gradients are constant per element for P_1 elements.

2.2 Biot–Savart law and its discretization

The magnetic flux density at sensor position $\mathbf{s}_m$ generated by the internal current density $\mathbf{J} = -\sigma \nabla u$ is

$$\mathbf{B}(\mathbf{s}_m) = \frac{\mu_0}{4\pi} \int_{\Omega} \mathbf{J}(\mathbf{r}) \times \frac{\mathbf{s}_m - \mathbf{r}}{|\mathbf{s}_m - \mathbf{r}|^3} \, dV. \quad (5)$$

With P_1 elements, ∇u is constant on each element and is obtained from the sparse gradient matrices $G_x, G_y, G_z \in \mathbb{R}^{n_{\text{el}} \times n_n}$, i.e. $(\partial_x u)_k = (G_x u)_k$, etc. The geometric part of the kernel is collected in the dense matrices $R_x, R_y, R_z \in \mathbb{R}^{M \times n_{\text{el}}}$,

$$(R_j)_{mk} = \int_{T_k} \frac{(\mathbf{s}_m - \mathbf{r})_j}{|\mathbf{s}_m - \mathbf{r}|^3} \, dV, \quad j \in \{x, y, z\}, \quad (6)$$

evaluated with a 35-point quadrature rule on each tetrahedron (`compute_r_matrices_local`, `dRmkj_dp`). Writing $\Sigma = \text{diag}(\sigma)$, expansion of the cross product in (5) yields the Cartesian components of the field at all sensors:

$$\begin{aligned} C_x(\sigma) &= -R_z \Sigma G_y + R_y \Sigma G_z, \\ C_y(\sigma) &= -R_x \Sigma G_z + R_z \Sigma G_x, \\ C_z(\sigma) &= -R_y \Sigma G_x + R_x \Sigma G_y, \end{aligned} \quad B_j(\mathbf{s}_m) = \frac{\mu_0}{4\pi} (C_j u)_m. \quad (7)$$

Each sensor m measures the projection of $\mathbf{B}$ onto its axes $\hat{\mathbf{g}}_d^{(m)}$, $d = 1, 2, 3$, stored as $g_{m,d,j}$. The measurement operator for axis d is

$$\Gamma_d(\sigma; \mathbf{s}) = \frac{\mu_0}{4\pi} (g_{:,d,x} \odot C_x + g_{:,d,y} \odot C_y + g_{:,d,z} \odot C_z) \in \mathbb{R}^{M \times n_n}, \quad (8)$$

where $\odot$ denotes row-wise scaling (`compute_gamma_matrices_local`). The measured datum for sensor m , stimulation l and axis d is then simply

$$B_{d,m}^{(l)} = (\Gamma_d u^{(l)})_m, \quad S(\sigma) u^{(l)} = b^{(l)}. \quad (9)$$

Note that Γ_d is *linear* in σ (through Σ) and depends on the sensor positions only through the matrices R_j in (6).

For *1-axis* MDEIT one fixed axis d is measured, giving MN data points; for *3-axis* MDEIT the data of all three axes are stacked, giving $3MN$ data points.

3 Bayesian inversion and the linearized posterior

Following the Bayesian paradigm, the discretized conductivity is a random vector with a Gaussian prior, and the data are corrupted by additive Gaussian noise,

$$\sigma \sim \mathcal{N}(\sigma_*, \Gamma_{\text{pr}}), \quad V = \mathcal{B}(\sigma; \mathbf{s}) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Gamma_{\text{noise}}), \quad (10)$$

where $\mathcal{B}$ stacks all measurements and $\mathbf{s} = (\mathbf{s}_1, \dots, \mathbf{s}_M)$ collects the design variables. To obtain a computationally tractable design criterion, the measurement map is linearized around the prior mean,

$$\mathcal{B}(\sigma) \approx \mathcal{B}(\sigma_*) + J(\sigma - \sigma_*), \quad J = J(\sigma_*; \mathbf{s}) = \left. \frac{\partial \mathcal{B}}{\partial \sigma} \right|_{\sigma_*}. \quad (11)$$

Under this linearization the posterior is Gaussian, with covariance

$$\Gamma_{\text{post}} = H^{-1}, \quad H = J^\top \Gamma_{\text{noise}}^{-1} J + \Gamma_{\text{pr}}^{-1}. \quad (12)$$

The essential observation for experiment design is that Γ_{post} *does not depend on the measured data* — only on the Jacobian J , hence on the sensor positions $\mathbf{s}$. The design can therefore be optimized offline, before any measurement is taken.

4 Optimality criteria

A-optimality. Minimizing the expected mean-square error of the conditional-mean estimate (with unit weighting) is equivalent to minimizing the trace of the posterior covariance:

$$\Phi_A(\mathbf{s}) = \text{tr}(\Gamma_{\text{post}}) = \text{tr}(H^{-1}) \quad (\text{compute_cost_function_a_opt}). \quad (13)$$

D-optimality. Maximizing the expected information gain (the Kullback–Leibler divergence from the prior to the posterior, marginalized over the data) is, for Gaussian linearized models, equivalent to minimizing the log-determinant of the posterior covariance:

$$\Phi_D(\mathbf{s}) = \log \det \Gamma_{\text{post}} = -\log \det H \quad (\text{compute_cost_function_d_opt}). \quad (14)$$

To avoid overflow/underflow, the determinant is evaluated through the Cholesky factorization $H = LL^\top$:

$$\log \det H = 2 \sum_i \log L_{ii}. \quad (15)$$

5 Gradients of the design criteria

Let p denote one Cartesian coordinate of one sensor position. Differentiating $H = J^\top \Gamma_{\text{noise}}^{-1} J + \Gamma_{\text{pr}}^{-1}$,

$$\frac{\partial H}{\partial p} = \frac{\partial J^\top}{\partial p} \Gamma_{\text{noise}}^{-1} J + J^\top \Gamma_{\text{noise}}^{-1} \frac{\partial J}{\partial p}. \quad (16)$$

Using $d \operatorname{tr}(H^{-1}) = -\operatorname{tr}(H^{-1} dH H^{-1})$ and $d \log \det H = \operatorname{tr}(H^{-1} dH)$, together with the symmetry of H , one obtains

$$\frac{\partial \Phi_A}{\partial p} = -2 \operatorname{tr} \left(H^{-2} J^\top \Gamma_{\text{noise}}^{-1} \frac{\partial J}{\partial p} \right) = -2 \left\langle W_A, \frac{\partial J}{\partial p} \right\rangle_F, \quad W_A = \Gamma_{\text{noise}}^{-1} J H^{-2}, \quad (17)$$

$$\frac{\partial \Phi_D}{\partial p} = -2 \operatorname{tr} \left(H^{-1} J^\top \Gamma_{\text{noise}}^{-1} \frac{\partial J}{\partial p} \right) = -2 \left\langle W_D, \frac{\partial J}{\partial p} \right\rangle_F, \quad W_D = \Gamma_{\text{noise}}^{-1} J H^{-1}, \quad (18)$$

where $\langle A, B \rangle_F = \sum_{ij} A_{ij} B_{ij}$ is the Frobenius inner product. The matrices H^{-1} and H^{-2} are computed by repeated triangular solves with the Cholesky factor L (functions `compute_cost_function_gradients`).

Since only the rows of J belonging to sensor m depend on $\mathbf{s}_m$, the Frobenius product in (17)–(18) reduces to a sum over that sensor's rows:

$$\frac{\partial \Phi}{\partial (\mathbf{s}_m)_p} = -2 \sum_{l=1}^N \sum_{k=1}^{n_{\text{el}}} W_{(m,l),k} \frac{\partial J_{(m,l),k}}{\partial (\mathbf{s}_m)_p}, \quad (19)$$

which is exactly the loop structure of the gradient routines. For 3-axis MDEIT the same expression is summed over the three axis blocks of the stacked Jacobian $J = [J_x; J_y; J_z]$.

6 The sensitivity Jacobian and its position derivatives

6.1 Jacobian with respect to conductivity (adjoint method)

The datum for sensor m , stimulation l (one fixed axis d ; the axis index is suppressed) is $B_m^{(l)} = (\Gamma u^{(l)})_m$ with $S(\sigma) u^{(l)} = b^{(l)}$. Differentiating with respect to σ_k and using $\partial u^{(l)} / \partial \sigma_k = -S^{-1}(\partial S / \partial \sigma_k) u^{(l)}$ gives

$$J_{(m,l),k} = \underbrace{\lambda_m^\top \frac{\partial S}{\partial \sigma_k} u^{(l)}}_{\text{adjoint part}} + \underbrace{\left(\frac{\partial \Gamma}{\partial \sigma_k} u^{(l)} \right)_m}_{\text{direct part}}, \quad S \lambda_m = -\Gamma^\top e_m, \quad (20)$$

where e_m is the m -th unit vector; the adjoint solves are performed with preconditioned conjugate gradients. By (4), the adjoint part is $|T_k| (\nabla \lambda_m)_k \cdot (\nabla u^{(l)})_k$, and since Γ is linear in σ , the direct part follows from (7) by replacing Σ with $e_k e_k^\top$. In the code these are the `dfdx` and `dfdp` terms of `calc_jacobian_1axis_direct_fully_vectorized_local`.

6.2 Position derivative of the Biot–Savart matrices

The only quantities depending on $\mathbf{s}_m$ are the rows of R_x, R_y, R_z . With $\mathbf{d} = \mathbf{s}_m - \mathbf{r}$, differentiating (6) under the integral sign gives

$$\frac{\partial (R_j)_{mk}}{\partial (\mathbf{s}_m)_p} = \int_{T_k} \frac{\delta_{jp} |\mathbf{d}|^2 - 3 d_j d_p}{|\mathbf{d}|^5} dV, \quad j, p \in \{x, y, z\}, \quad (21)$$

evaluated with the same quadrature rule (`dRmkj_dp`, `dRmkj_xyz`). This explicit, smooth derivative is what replaces the electrode shape derivatives of the EIT setting.

6.3 Position derivative of the Jacobian

Differentiating $J_{(m,l),k}$ with respect to $p = (\mathbf{s}_m)_p$ produces two contributions (functions `compute_dJ_ds`, `compute_dJ_xyz`, `compute_dJxyz_xyz`):

$$\frac{\partial J_{(m,l),k}}{\partial p} = \underbrace{|T_k| \left(\nabla \frac{\partial \lambda_m}{\partial p} \right)_k \cdot (\nabla u^{(l)})_k}_{\text{dJ1}} + \underbrace{\frac{\partial}{\partial p} \left(\frac{\partial \Gamma}{\partial \sigma_k} u^{(l)} \right)_m}_{-\text{dJ2}}. \quad (22)$$

- The second term is obtained directly from (7) and (21): it is the direct part of the Jacobian with R_j replaced by $\partial R_j / \partial p$.
- The first term requires the derivative of the adjoint solutions. Since S does not depend on the sensor positions, differentiating $S \lambda_m = -\Gamma^\top e_m$ gives one extra adjoint solve per sensor and coordinate,

$$S \frac{\partial \lambda_m}{\partial p} = -\frac{\partial \Gamma^\top}{\partial p} e_m, \quad (23)$$

where $\partial \Gamma / \partial p$ again follows from (8) and (21) (functions `compute_dlambdas`, `compute_dlambdas_xyz`, `compute_dlambda_xyz`).

Note that the forward solutions $u^{(l)}$ do not depend on the sensor positions at all — another consequence of the electrodes being fixed.

7 Parametrization, constraints and optimization

7.1 Cylindrical coordinates and chain rule

For a cylindrical tank it is convenient to parametrize each sensor by $q_m = (r_m, \theta_m, z_m)$ with $\mathbf{x}_m = (r_m \cos \theta_m, r_m \sin \theta_m, z_m)$. The gradient with respect to q follows from the chain rule,

$$\frac{\partial \Phi}{\partial q_i} = \sum_{p=1}^3 \frac{\partial x_p}{\partial q_i} \frac{\partial \Phi}{\partial x_p}, \quad \frac{\partial \mathbf{x}}{\partial (r, \theta, z)} = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -r \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (24)$$

implemented in `dphidp_to_dphidq` with the transformation matrix from `compute_jacobian_coordinate_trans`

Remark 1. The transformation matrix in (24) depends on the current sensor coordinates (r_m, θ_m) and must therefore be re-evaluated at every iterate of the optimizer, not only at the initial configuration.

Two constrained search domains are used with the interior-point algorithm of `fmincon` (analytic gradients supplied, L-BFGS Hessian approximation):

- *cylinder shell*: $r_m = r_0$ fixed, θ_m free, $z_m \in [z_{\min}, z_{\max}]$;
- *annular region*: $r_m \in [r_{\min}, r_{\max}]$, θ_m free, $z_m \in [z_{\min}, z_{\max}]$.

For unconstrained optimizers, bounded coordinates are mapped to $\mathbb{R}$ by the logistic reparametrization $r = r_{\min} + (r_{\max} - r_{\min}) \varsigma(\xi)$, $\varsigma(\xi) = (1 + e^{-\xi})^{-1}$, and analogously for z (`map_x_to_q_cyl_region` / `map_q_to_x_cyl_region`).

7.2 Sensor repulsion penalty

Analogous to the electrode-gap penalty of Hyvönen et al., a smooth pairwise repulsion term can be added to prevent sensor clustering:

$$\Phi_{\text{rep}}(\mathbf{s}) = \frac{1}{2} \sum_{i \neq j} \frac{1}{|\mathbf{x}_i - \mathbf{x}_j|^2 + \epsilon}, \quad \frac{\partial \Phi_{\text{rep}}}{\partial \mathbf{x}_i} = -2 \sum_{j \neq i} \frac{\mathbf{x}_i - \mathbf{x}_j}{(|\mathbf{x}_i - \mathbf{x}_j|^2 + \epsilon)^2}, \quad (25)$$

with a small $\epsilon > 0$ for smoothness (`repulsion_cost`, `repulsion_grad_cartesian`). The total objective is $\Phi = \Phi_{\text{OED}} + \alpha \Phi_{\text{rep}}$ with a manually chosen weight $\alpha \geq 0$.

8 Practical choices of the noise and prior covariances

The script uses white (uncorrelated) covariances,

$$\Gamma_{\text{noise}} = \beta I, \quad \Gamma_{\text{pr}} = \alpha_{\text{pr}} I, \quad (26)$$

scaled as follows:

- The noise standard deviation is tied to the magnitude of the simulated data, $\sqrt{\beta} = \max_{j,m,l} |B_{j,m}^{(l)}| / 10$.
- The prior variance is set relative to the largest eigenvalue of $J_0^\top J_0$ (the Jacobian at the initial configuration),

$$\alpha_{\text{pr}} = c \frac{\beta}{\lambda_{\max}(J_0^\top J_0)}, \quad c = 50. \quad (27)$$

The ratio α_{pr}/β controls how many eigenmodes of the problem are *data-dominated*: the number of eigenvalues λ_i of $J_0^\top J_0$ with $\lambda_i \alpha_{\text{pr}}/\beta > 1$ counts the modes for which the (linearized) data constrain the posterior more strongly than the prior. If this count is very small, the design criteria are dominated by the prior and become insensitive to the sensor positions; if it is very large, the linearization around the prior mean may become the limiting approximation.

9 Summary of the algorithm

1. Build the FE model, fix electrodes, stimulation patterns, Γ_{pr} , Γ_{noise} ; draw an initial sensor configuration $\mathbf{s}^{(0)}$.
2. At the current configuration: assemble $R_j(\mathbf{s})$, $\Gamma(\sigma_*; \mathbf{s})$; solve the forward problems $Su^{(l)} = b^{(l)}$ and adjoint problems $S\lambda_m = -\Gamma^\top e_m$; form the Jacobian J .
3. Form $H = J^\top \Gamma_{\text{noise}}^{-1} J + \Gamma_{\text{pr}}^{-1}$, its Cholesky factor, and evaluate $\Phi_A = \text{tr}(H^{-1})$ or $\Phi_D = -\log \det H$.
4. Compute $\partial R_j / \partial p$ from (21), the extra adjoint solves for $\partial \lambda_m / \partial p$, and assemble $\partial J / \partial p$ from (22); contract with W_A or W_D as in (17)–(18); transform the gradient to the optimization coordinates via (24).
5. Feed cost and gradient to the constrained optimizer (`fmincon`, interior-point, L-BFGS); iterate until convergence.

10 Case study: sensors confined to a circle at fixed height

This section documents a numerical case study, implemented in the sibling script `example_anomaly_circle.m`, that specializes the theory above to the simplest non-trivial circle-constrained design: M magnetometers on a circle of *fixed* radius $r_m \equiv r_s$ and *fixed* height $z_m \equiv z_s$ (half the tank height), so that only the azimuths θ_m are free. Three-axis MDEIT is used throughout, $J = [J_x; J_y; J_z]$, and the cylindrical chain rule (24) collapses to the one-dimensional form

$$\frac{\partial \Phi}{\partial \theta_m} = -r_s \sin \theta_m \frac{\partial \Phi}{\partial x_m} + r_s \cos \theta_m \frac{\partial \Phi}{\partial y_m}, \quad (28)$$

evaluated at the *current* θ_m at every iterate (cf. the remark in §8 above about the analogous (r, θ) dependence).

10.1 Setup: ROI-informative prior and target anomaly

A single conductive spherical anomaly (radius $R/3$, conductivity $5 \times$ background) is placed off-centre at half height, at $c_0 = (R/2, 0, H/2)$, where R, H are the tank radius and height. The prior is informative about *where* the anomaly may be but not its value, following the “Case 1” prior of Hyvönen et al.:

$$(\Gamma_{\text{pr}})_{kk} = \begin{cases} \sigma_{\text{roi}}^2, & \text{centroid}(T_k) \in \mathcal{R}, \\ \sigma_{\text{bg}}^2, & \text{otherwise,} \end{cases} \quad \mathcal{R} = \{k : |\text{centroid}(T_k) - c_0| \leq \rho\}, \quad (29)$$

with ρ a multiple (`roi_radius_factor`) of the anomaly radius. Sensors are confined to $r_m = r_s = 1.5R$, $z_m = z_s = H/2$ ($M = 8$), so (28) is the only coordinate transformation needed.

10.2 Noise calibration by target data-dominated rank

Rather than the fixed $\sqrt{\beta} = \max |B|/10$ rule of §8, the case study calibrates the noise level directly from the whitened-Jacobian spectrum at the even-spacing configuration $\mathbf{s}_0$. Let $J_0 = J(\sigma_*; \mathbf{s}_0)$ and let $s_1 \geq s_2 \geq \dots$ be the singular values of $J_0 \Gamma_{\text{pr}}^{1/2}$. Setting

$$\sqrt{\beta} = s_d \quad (30)$$

for a target rank d (`d_target`) makes exactly d modes data-dominated ($s_i^2/\beta > 1$), placing the design problem in the regime where the sensor positions can actually matter: too much noise and the criteria are prior-dominated and design-insensitive; too little and every configuration already saturates them (cf. the discussion in §8). A companion diagnostic, the mean ROI energy of the leading right singular vectors $v_1, \dots, v_d$,

$$\mathcal{E}_{\mathcal{R}} = \frac{1}{d} \sum_{i=1}^d \sum_{k \in \mathcal{R}} (v_i)_k^2, \quad (31)$$

measures how much of the data-dominated subspace actually overlaps the ROI, as opposed to the much larger and better-conditioned background.

10.3 Numerical results

With $\sigma_{\text{bg}} = 0.005 \sigma_0$, $\sigma_{\text{roi}} = \sigma_0$ (σ_0 the background conductivity), ρ equal to one anomaly radius, and $d = 100$ ($n_{\text{el}} = 3425$, $n_{\text{data}} = 384$), the even-spacing configuration already reduces the ROI posterior variance by about 30% relative to the prior ($\mathcal{E}_{\mathcal{R}} \approx 0.54$). Optimizing from this starting point (40 quasi-Newton iterations of `fminunc`, analytic gradient, `SpecifyObjectiveGradient`) gives:

Criterion	$\Phi(\text{even})$	$\Phi(\text{optimized})$	Gain
A-optimality, $\text{tr}(H^{-1})$	2.879×10^{-1}	2.747×10^{-1}	4.6% trace reduction
D-optimality, $-\log \det H$	-5.555×10^4	-5.557×10^4	17.6 nats (25.4 bits)

Both criteria move the sensors toward the anomaly’s azimuth, visibly more so for D-optimality. A multistart check (3 random initial azimuth sets) converges to the same A-optimal value in every run, and to within 0.03% of the same D-optimal value, confirming that the even-spacing start is already close to the global optimum of this low-dimensional ($M = 8$) design space.

10.4 Discussion: why the A-optimal gain stays modest

Remark 2. *Enlarging the data budget alone does not close the gap between the two criteria: repeating the optimization for $d \in \{60, 100, 150\}$ leaves the converged A-optimal trace reduction in the same 4–6% range throughout, which rules out an insufficient-SNR explanation. Moving the anomaly closer to the tank wall (from $R/2$ to $0.6R$, all else unchanged) leaves the (3-iterate) A-optimal reduction almost unchanged (2.7% $\rightarrow$ 4.4%) while nearly doubling the D-optimal information gain (13.8 $\rightarrow$ 24.8 nats). The bottleneck is therefore geometric, not statistical.*

A-optimality is an *average* of marginal posterior variances, $\Phi_A = \sum_k (\Gamma_{\text{post}})_{kk}$, and is comparatively insensitive to *correlations* between sensor measurements: two sensors whose Jacobian rows are nearly collinear (typical for evenly spaced sensors seeing a common, roughly axisymmetric electrode/stimulation pattern) each still contribute their own, largely redundant, reduction to the trace. D-optimality, $\Phi_D = -\log \det H = -\sum_i \log \lambda_i(H)$, behaves oppositely: an eigenvalue of H associated with a direction seen by two collinear sensors stays pinned near its prior value, so *removing* that redundancy by reallocating sensors is disproportionately rewarded. This is consistent with the classical design-of-experiments folklore that A-optimal designs tolerate point replication far better than D-optimal designs. Because the fixed-radius, fixed-height circle used here is itself a highly symmetric search domain, the even-spacing configuration is close to a “replicated” design in this sense: moving the sensors mainly re-shuffles *which* directions are well resolved (a large D-optimal effect) rather than *how much* variance is removed *on average* (a small A-optimal effect). We have not proved this as a theorem; it is offered as the most parsimonious explanation consistent with all of the numerical evidence above. A natural follow-up, left to future work, is whether relaxing the height constraint — letting z_m vary together with θ_m , still at fixed radius r_s — breaks enough of this residual symmetry to produce a more considerable A-optimal gain.

10.5 Follow-up: freeing the height z_m

The question closing the previous subsection is answered here using a sibling script, `example_anomaly_circle_t`, identical to `example_anomaly_circle.m` in every other setting (anomaly, prior, noise calibration, $d = 100$, 3-axis MDEIT, `fminunc` quasi-Newton with analytic gradient) except that each sensor now has *two* free coordinates (θ_m, z_m) at the same fixed radius $r_s = 1.5R$. To keep the unconstrained solver `fminunc` usable with a single analytic gradient, z_m is generated from an unconstrained η_m through a sigmoid,

$$z_m = z_{\min} + (z_{\max} - z_{\min}) \text{sigm}(\eta_m), \quad \text{sigm}(y) = \frac{1}{1 + e^{-y}}, \quad (32)$$

with $z_{\min} = 0$, $z_{\max} = H$, so that $\eta_m = 0$ reproduces the previous fixed height $z_s = H/2$ exactly and $q_0 = (\theta_{\text{even}}, \mathbf{0})$ is an identical-to-baseline start. The chain rule (28) for θ_m is unchanged; height contributes the new term

$$\frac{\partial \Phi}{\partial \eta_m} = \frac{\partial \Phi}{\partial z_m} (z_{\max} - z_{\min}) \text{sigm}(\eta_m) (1 - \text{sigm}(\eta_m)), \quad (33)$$

reusing the $p = 3$ (Cartesian z) component of $\partial\Phi/\partial\mathbf{p}_m$ that the fixed-height script already computed and discarded. Both halves of the resulting $2M$ -length gradient were verified against central finite differences (rel. err. 10^{-6} – 10^{-8} , the same standard used throughout this study).

Results. Optimizing from q_0 (40 quasi-Newton iterations, as in the theta-only case) gives:

Criterion	$\Phi(\text{even})$	$\Phi(\text{optimized})$	Gain
A-optimality, $\text{tr}(H^{-1})$	2.879×10^{-1}	2.601×10^{-1}	9.7% trace reduction
D-optimality, $-\log \det H$	-5.555×10^4	-5.558×10^4	29.0 nats (41.9 bits)

A longer single-start run (150 iterations) confirms both optima are fully converged (A-optimal gradient norm 7.6×10^{-9} ; D-optimal run halts because it cannot decrease further along the search direction, gradient norm 2.4×10^{-5}) at essentially the same values (9.7%, 29.0 nats), so the 40-iteration figures above are not an artifact of premature stopping. A multistart check (4 random initial (θ, η) sets, 60 iterations each) finds every run within 0.3% of the same A-optimal value and within 0.003% of the same D-optimal value (best of the four: 9.9% trace reduction, 30.3 nats), confirming that — as in the theta-only case — the even-spacing start is close to the global optimum of this now $2M = 16$ -dimensional design space and that no qualitatively better configuration is being missed.

Diagnosis. Freeing z_m roughly *doubles* both gains relative to the theta-only results of §10 (4.6% $\rightarrow$ 9.7–9.9% for A-optimality, $17.6 \rightarrow 29$ –30 nats for D-optimality), which is a real and reproducible effect, not noise: a non-trivial part of the modest A-optimal gain in §10 *was* a symmetry artifact of the overly restrictive fixed-radius-fixed-height search domain, exactly as conjectured. However, the new A-optimal reduction ($\approx 10\%$) still falls short of being “considerable” by the same standard applied throughout this study ($\gtrsim 15$ –30%), so the restrictive-domain hypothesis is only *partially* confirmed. Inspecting the optimized (θ_m, z_m) pairs shows why: sensors still cluster in azimuth toward the anomaly’s side (as before), but in height they *spread out* across a substantial fraction of the shell ($\text{std}(z_m) \approx 0.36$ – 0.40 m for A-optimality, ≈ 0.21 m for D-optimality, out of a 1.75 m range) rather than converging tightly onto the anomaly’s own height $z = H/2$. This is the same decorrelation/de-clustering mechanism identified in §10 for azimuth, now acting in the height direction as well: spreading in z reduces redundancy between sensor rows (helping both criteria, but disproportionately D-optimality) rather than concentrating averaged sensitivity near the anomaly (which is what would move the A-optimal trace substantially). In short: the fixed-height restriction explains *part*, but evidently not *all*, of the gap between the two criteria’s gains — the remainder is consistent with the more fundamental explanation from §10, that A-optimality (an averaged criterion) is intrinsically more tolerant of replicated/correlated sensor information than D-optimality, independent of how many geometric degrees of freedom the design domain provides.

11 Case study: human thorax, sensors on a fixed circle at half height

The two cylinder studies above used a rotationally symmetric conductor. This section repeats the azimuth-only design (§10) on a markedly more complicated geometry: an adult-male thorax cross-section from the EIDORS `shape_library`, extruded to a 3D slab of height $H = 0.5$, with the two lungs as interior low-conductivity regions ($\sigma_{\text{bg}} = 1$, $\sigma_{\text{lung}} = 0.3$; this homogeneous-lung image is the OED linearization point / prior mean). Twelve CEM electrodes form one ring at half height; $M = 16$ magnetometers lie on a circle of *fixed* radius $R_0 = 1.5$ at $z_s = H/2$, only their azimuths θ_m free, so (28) is again the only coordinate transformation. The implementation is `human_thorax_theta.m`; it drives the shared, finite-difference-validated optimizer

`optimize_sensor_configuration.m` with the theta coordinate maps, and the analytic $\partial\Phi/\partial\theta_m$ was re-verified against central finite differences (rel. err. 10^{-9} – 10^{-6}) at both the even start and a random perturbed configuration, for A- and D-optimality.

Prior, noise, baselines. The diagonal “Case 1” prior (29) is used, with the suspected-anomaly centre $c_0 = (0.1, 0.1, H/2)$, $\sigma_{\text{bg}} = 0.005$ and $\sigma_{\text{roi}} = 1$. On the coarse thorax mesh ($n_{\text{el}} = 5955$) a factor-one ROI ball holds only 14 elements, making the ROI-energy diagnostic (31) pathologically small ($\mathcal{E}_{\mathcal{R}} \approx 0.11$); enlarging the ball to $\rho = 2\times$ the anomaly radius restores a 129-element ROI (comparable to the cylinder’s ~ 100) with $\mathcal{E}_{\mathcal{R}} \approx 0.33$. The noise is calibrated by (30) with $d = 100$ ($n_{\text{data}} = 576$), giving $\sqrt{\beta} \approx 1.4 \times 10^{-6}$; at this level the even configuration already reduces the ROI posterior variance by about 26% relative to the prior. Two baselines are compared against, both at $M = 16$: (i) even azimuthal spacing (also the optimizer start), and (ii) a “boundary” configuration with sensors at a fixed standoff $\delta = 0.05$ from the body along equiangular rays — the thorax analogue of “as close to the conductor as allowed”.

Results. Optimizing from the even start (40 quasi-Newton iterations):

Criterion	$\Phi(\text{even})$	$\Phi(\text{optimized})$	Gain vs. even
A-optimality, $\text{tr}(H^{-1})$	9.5559×10^1	9.5415×10^1	0.15% trace reduction
D-optimality, $-\log \det H$	-6.20443×10^4	-6.20482×10^4	3.9 nats (5.6 bits)

The A-optimal even start reaches “Local minimum found”; a multistart check (three random initial azimuth sets) agrees to $\leq 0.003\%$ on the A-optimal value and to all printed digits on the D-optimal value, so these are genuine converged optima, not premature stops. Crucially, the “as close as allowed” boundary configuration is *far* better than any fixed- R_0 circle: $\Phi_{\text{A}} = 71.08$ (a further 25% below the azimuth optimum) and $\Phi_{\text{D}} = -6.312 \times 10^4$ (a further 1076 nats), consistent with the $1/r^2$ Biot–Savart decay — proximity, not azimuth, is the dominant lever, and it is entirely unavailable at fixed R_0 .

Discussion: weaker, not stronger, than the cylinder. The a priori expectation was that the thorax, breaking rotational symmetry twice (non-circular boundary; off-centre ROI), would yield a *larger* A-optimal azimuth gain than the cylinder’s 4.6%. The opposite occurs (0.15%), and the reason is geometric and quantifiable. What the averaged A-optimal trace responds to is the azimuthal *contrast* in the ring’s sensitivity to the ROI, i.e. the spread of ROI-to-sensor distances around the circle. For the specified centre c_0 (radius $|c_0| \approx 0.14$ from the axis) against $R_0 = 1.5$, that distance varies only by $\pm 9.4\%$ around the ring; the cylinder’s anomaly at $0.5R$ against a $1.5R$ ring varied by $\pm 32.5\%$. The ROI here is in effect *more central* relative to the sensor circle than the cylinder anomaly was, so the fixed-radius circle conserves the average sensitivity budget under azimuth moves even more nearly than in §10. The body’s non-circular boundary does not rescue this: with the sensors held far out at $R_0 = 1.5$, the anomaly signal δB is governed by the sensor-to-ROI distance, not by the (strongly azimuth-dependent) sensor-to-surface standoff, so the boundary shape barely enters the ROI sensitivity. The optimized azimuths are nonetheless still weakly *structured* in the expected direction — for both criteria the number of sensors within $\pm 45^\circ$ of the ROI azimuth (45°) rises from 3 to 5, with sensors bunching on the ROI/near-body (north-east) side and thinning on the far side (A-optimality: 9 of the 16 sensors end up in the 0 – 144° arc; largest single-sensor move $\approx 36^\circ$) — by the same redundancy-reduction mechanism as before, and D-optimality again shows the larger relative effect. But on this near-central-ROI, far-ring design the resultant clustering is mild ($|\sum e^{i\theta_m}|/M \approx 0.17$) and the absolute gains are small for both criteria. The lesson carried to the radius-free study (Task B) is that on the thorax the decisive degree of freedom is radial standoff, not azimuth: the boundary configuration’s large advantage is entirely a proximity effect that only a free (or noise-traded) radius can access.

Symbol–code correspondence

Symbol / quantity	Code
$S(\sigma)$, eq. (3)	<code>M(img,sigma)</code>
R_x, R_y, R_z , eq. (6)	<code>compute_r_matrices_local</code>
Γ_d , eq. (8)	<code>compute_gamma_matrices_local (img.Gamma1..3)</code>
J (adjoint + direct part)	<code>calc_jacobian_laxis_direct_fully_vectorized_local</code>
λ_m	<code>compute_lambda_x</code> / PCG solves inside the Jacobian routine
$\partial R_j / \partial p$, eq. (21)	<code>dRmkj_dp, dRmkj_xyz</code>
$\partial \lambda_m / \partial p$	<code>compute_dlambda_ds, compute_dlambda_xyz</code>
$\partial J / \partial p$, eq. (22)	<code>compute_dJ_ds, compute_dJ_xyz, compute_dJxyz_xyz</code>
Φ_A, Φ_D	<code>compute_cost_function_{a,d}_opt(_3_axis)</code>
W_A, W_D , eqs. (17)–(18)	<code>compute_cost_function_gradient_{a,d}_opt_optimized(_3_axis)</code>
Chain rule, eq. (24)	<code>dphidp_to_dphidq</code>
Φ_{rep}	<code>repulsion_cost, repulsion_grad_cartesian</code>

Reference

N. Hyvönen, A. Seppänen, S. Staboulis, *Optimizing electrode positions in electrical impedance tomography*, SIAM Journal on Applied Mathematics 74(6), 1831–1851 (2014). [arXiv:1404.7300](#).